

Network design practice 4-2

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1 Assignment 1

```
cellulargbuntu:~/UERANSIM$ cd config/
cellulargbuntu:~/UERANSIM/config$ cat custom-gnb.yaml
cat: custom-gnb.yaml: No such file or directory
cellulargbuntu:~/UERANSIM/config$ ls -a
.  ..  2022171063-gnb.yaml  2022171063-ue.yaml  ue.rlp
cellulargbuntu:~/UERANSIM/config$ cd ..
cellulargbuntu:~/UERANSIM$ cd build/
cellulargbuntu:~/UERANSIM/build$ ./nr-gnb -c ../config/custom-gn
p.yaml
UERANSIM v3.2.7
2026-05-25 18:51:43.550 [sctp] [info] Trying to establish SCTP
connection... (192.168.48.131:38412)
2026-05-25 18:51:43.589 [sctp] [info] SCTP connection establish
ed (192.168.48.131:38412)
2026-05-25 18:51:43.600 [sctp] [debug] SCTP association setup a
cellulargbuntu:~/UERANSIM/build$ ./nr-ctl ins1-999782022171063
-exec 'ps -lsth'
PDU Session1:
state: PS-ACTIVE
session-type: IPv4
apn: Internet
s-mssai:
sst: 0x01
sd: null
emergency: false
address: 10.45.0.2
ambr: up[1000000Kb/s] down[1000000Kb/s]
data-pending: false
```

Figure 1: 1Gps DL/UL AMBR

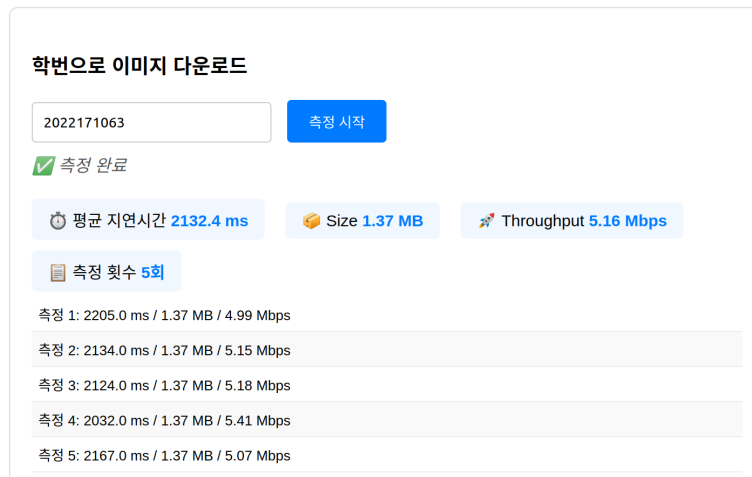


Figure 2: Downloaded image result

Observed clean download with 5.16Mbps throughput. Latency is 2132ms.

2 Assignment 2



Figure 3: Adjusted AMBR rate(top) Downloaded image result(bottom)

Adjusted AMBR DL/UL rate to 1Mbps. The result shows that the throughput is now 0.84Mbps.

Compared to the throughput of the previous practice where the AMBR was 1Gbps, 5.16Mbps, it is now capped at 1Mbps. This leads to lower throughput of 0.84Mbps with a 13014ms latency.

The transmission speed is being bottlenecked by various environmental capabilities. AMBR is merely indicating the maximum ceiling transmission rate. Therefore the actual throughput is lower than the set AMBR.